Can one finite switch change the reflected wave?

Illustrative target: about 180° between states, with useful reflected magnitude.



$L = a + d$

L=0.360 a=0.110 d=0.250

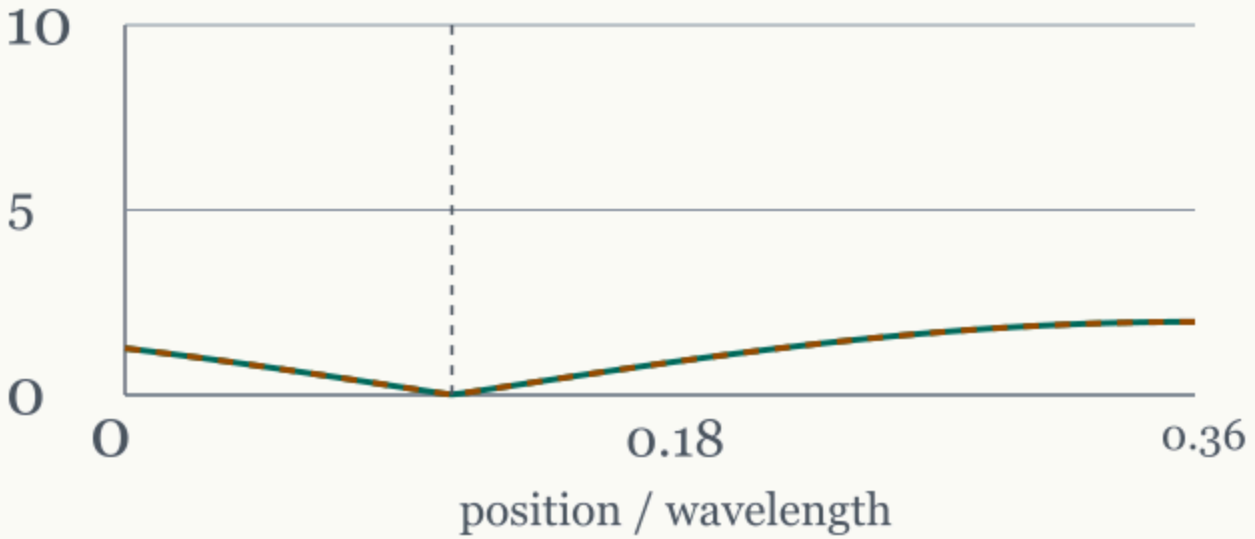
Port reflection

ON

OFF



Physical current magnitude



0.026°

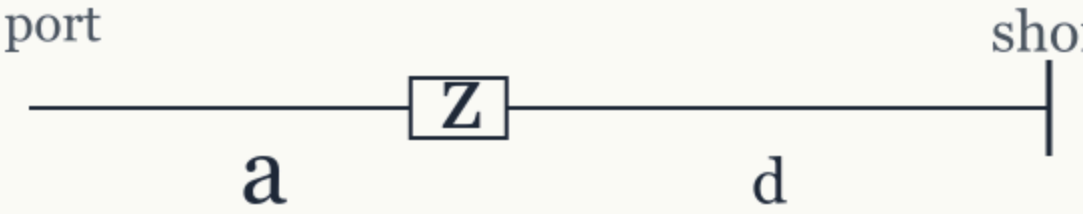
phase separation

min |R| = 0.9786

Same switch, same short

Can one finite switch change the reflected wave?

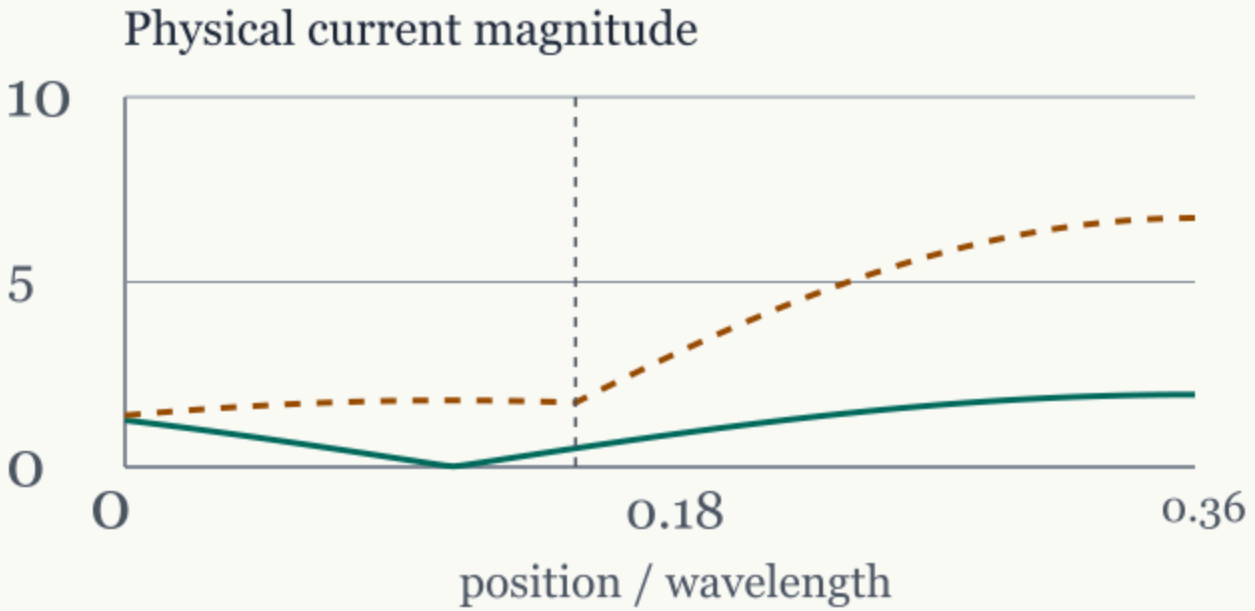
Illustrative target: about 180° between states, with useful reflected magnitude.



$L = a + d$

L=0.360 a=0.152 d=0.208





179.686°

phase separation

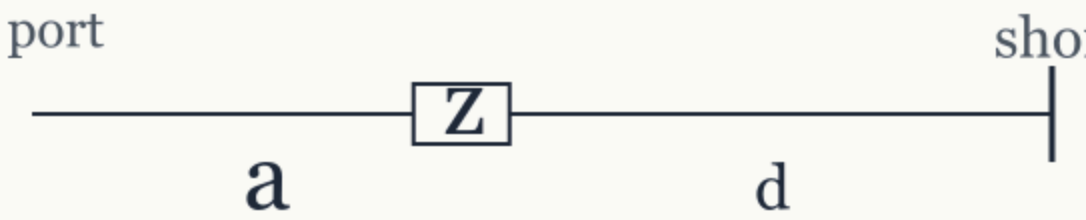
min |R| = 0.8009

Same switch, same short

The solved switch response changes when current participates.

What did the minimal case keep?

A synthetic causal analogue. It is not calibrated to the researcher's 3D copper unit cell.



$L = a + d$

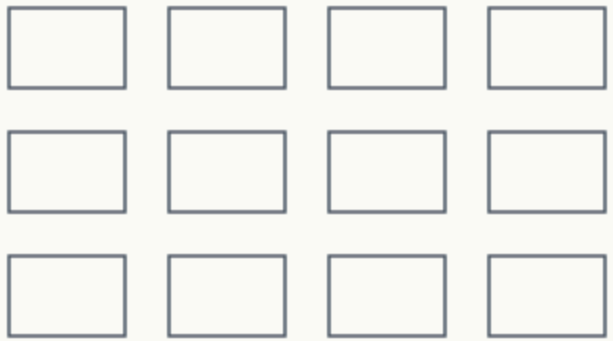
L=0.360 a=0.152 d=0.208

Port reflection

ON

OFF





Conceptual periodic array

$$\tilde{V}(0) + (1 + z_s)\tilde{I}(0) = 2, \qquad \tilde{V}(1) = 0$$

Finite switch: voltage jump = z I

Common termination in both states

What did the minimal case keep?

A synthetic causal analogue. It is not calibrated to the researcher's 3D copper unit cell.



$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



$$\tilde{V}(0) + (1 + z_s)\tilde{I}(0) = 2, \qquad \tilde{V}(1) = 0$$

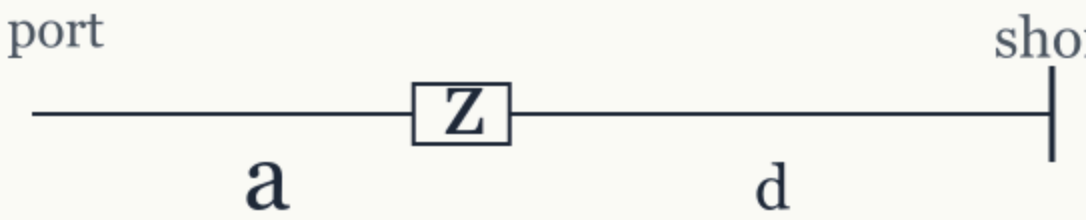
Finite switch: voltage jump = z I

Common termination in both states

Conceptual periodic array

What did the minimal case keep?

A synthetic causal analogue. It is not calibrated to the researcher's 3D copper unit cell.



L = a + d

L=0.360 a=0.152 d=0.208



$$\xi = \frac{x-a}{d}, \quad \tilde{V}_\xi + \gamma d \tilde{I} = 0, \quad \tilde{I}_\xi + \gamma d \tilde{V} = 0$$

$$\tilde{V}(0) + (1 + z_s)\tilde{I}(0) = 2, \qquad \tilde{V}(1) = 0$$

Finite loading survives. Radiation, materials and hardware mapping need validation.

What must the derivative equal at this point?

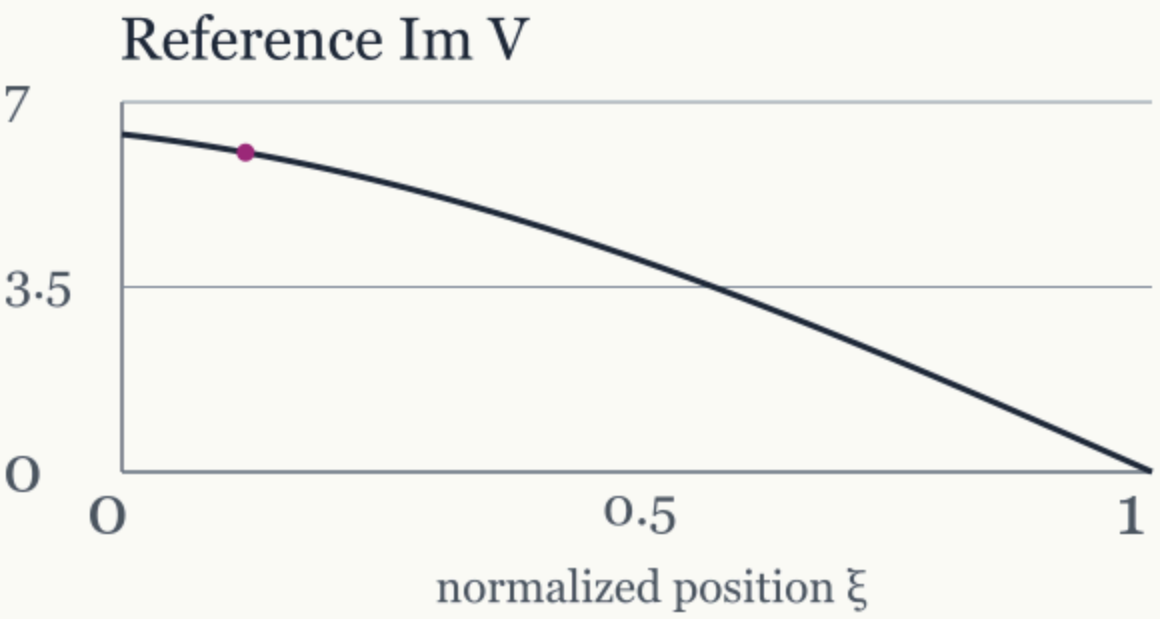
Harmonic PDE reduction: $V\xi + \gamma d\, I = 0$. These are complex amplitudes, not a transient.

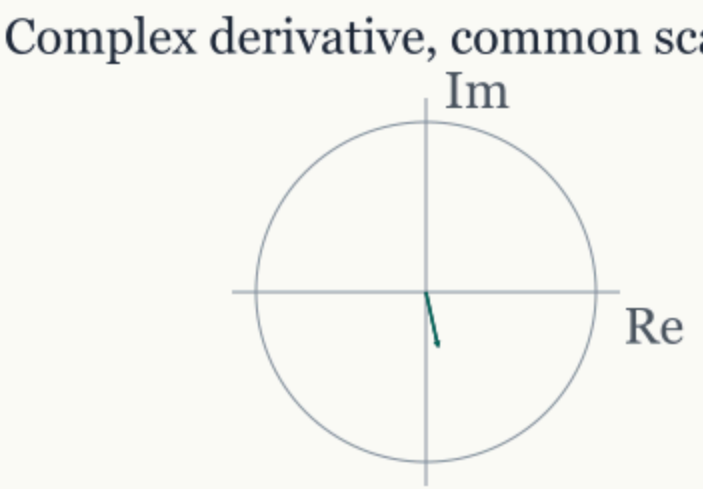


$L = a + d$

L=0.360 a=0.152 d=0.208







FD derivative: +0.776 -3.513i

Required $-\gamma d\, I$: +0.776 -3.513i

ξ = 0.12, independent wave-voltage FD

What must the derivative equal at this point?

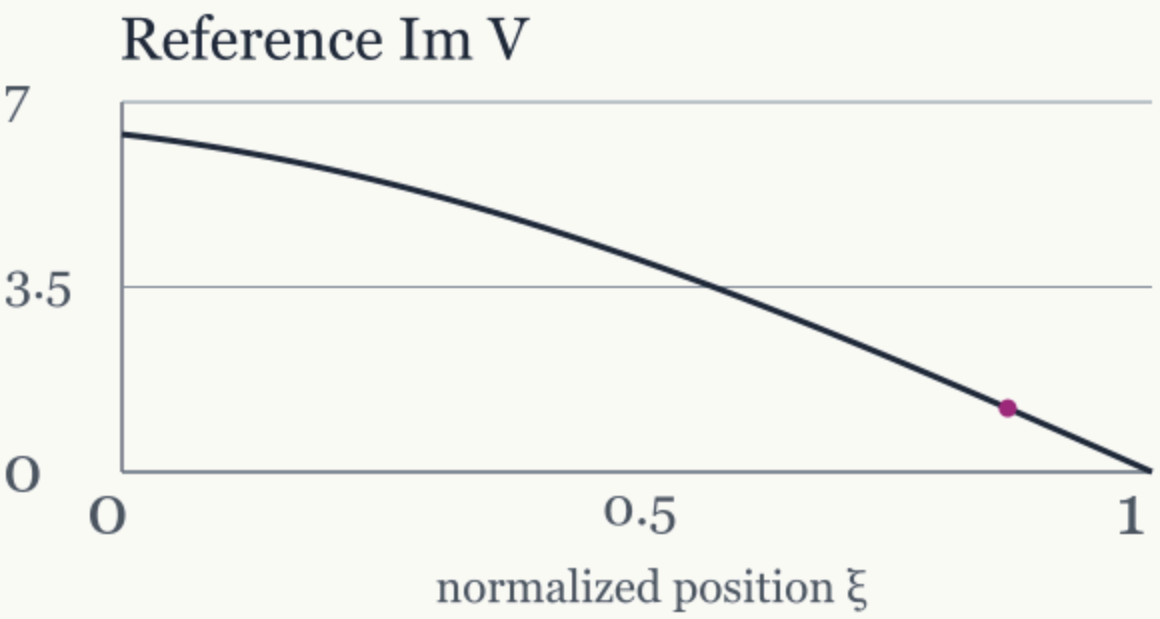
Harmonic PDE reduction: $V\xi + \gamma d\, I = 0$. These are complex amplitudes, not a transient.

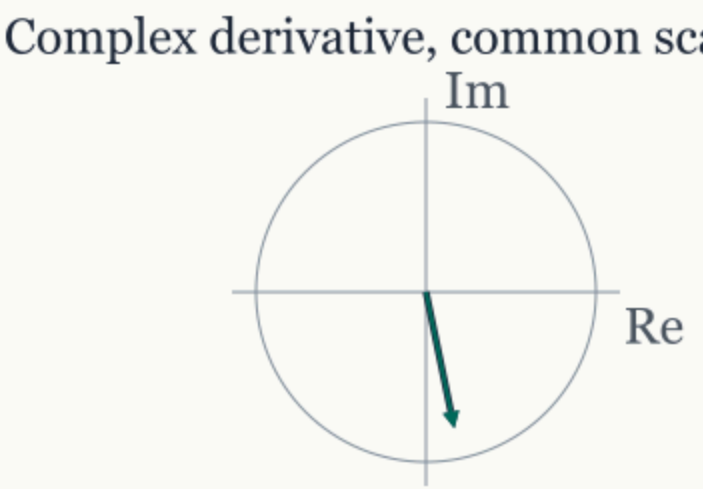


$L = a + d$

L=0.360 a=0.152 d=0.208







FD derivative: +1.772 -8.520i

Required $-\gamma d\, I$: +1.772 -8.520i

ξ = 0.86, independent wave-voltage FD

Agreement at one point does not establish the whole field or its boundaries.

Which change controls phase separation?

Move the switch at fixed L, then vary upstream a while holding right length d fixed.



$L = a + d$

L=0.360 a=0.120 d=0.240

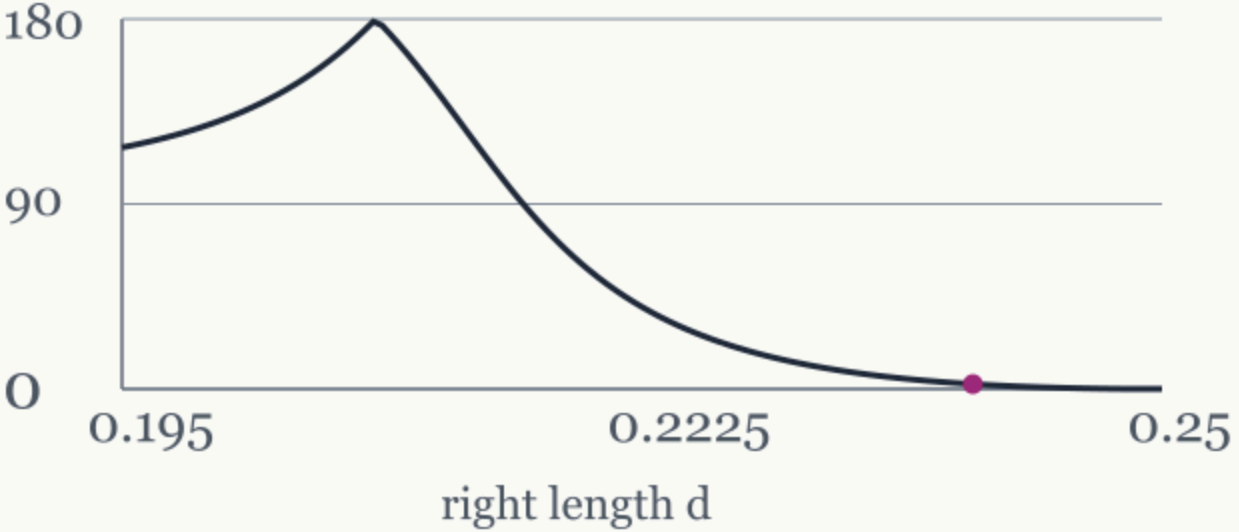
Port reflection

ON

OFF



Exact phase separation



$$R_s = e^{-2\gamma a}\frac{\sinh(\gamma d) + (z_s - 1)\cosh(\gamma d)}{\sinh(\gamma d) + (z_s + 1)\cosh(\gamma d)}$$

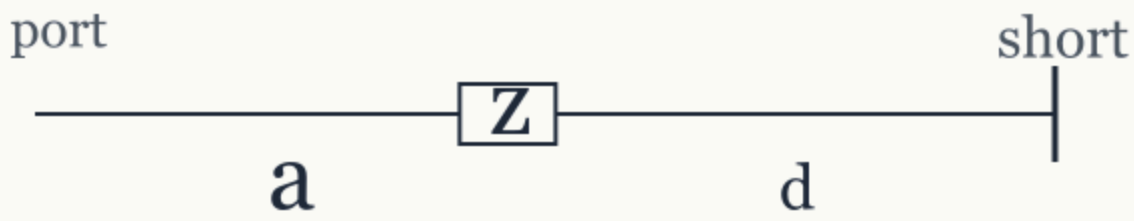
Output-residual gain G = 1.855

$\Delta\varphi = 2.371°$

The common factor rotates and attenuates.

Which change controls phase separation?

Move the switch at fixed L, then vary upstream a while holding right length d fixed.



$L = a + d$

L=0.388 a=0.180 d=0.208

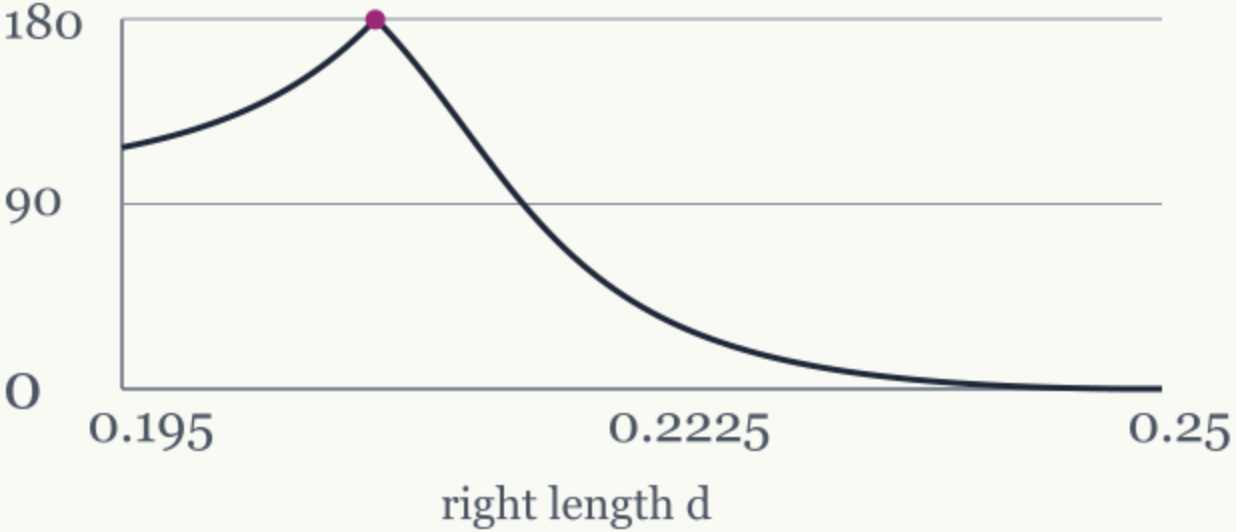
Port reflection

ON

OFF



Exact phase separation



$$R_s = e^{-2\gamma a}\frac{\sinh(\gamma d) + (z_s - 1)\cosh(\gamma d)}{\sinh(\gamma d) + (z_s + 1)\cosh(\gamma d)}$$

Output-residual gain G = 5.035

$\Delta\varphi = 179.686°$

The common factor rotates and attenuates.

Which change controls phase separation?

Move the switch at fixed L, then vary upstream a while holding right length d fixed.



$L = a + d$

L=0.360 a=0.152 d=0.208

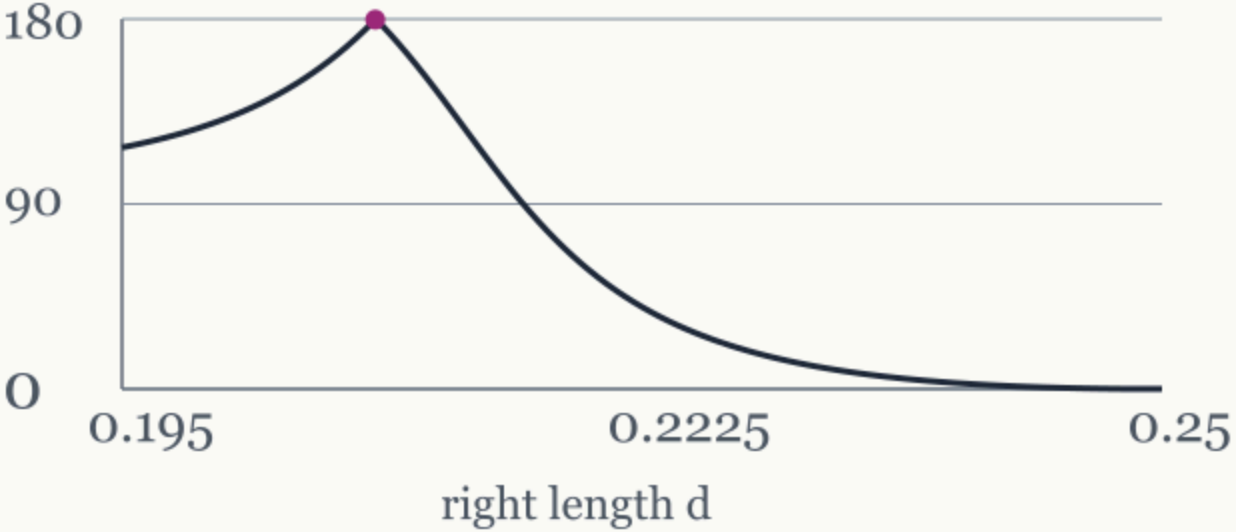
Port reflection

ON

OFF



Exact phase separation



$$R_s = e^{-2\gamma a}\frac{\sinh(\gamma d) + (z_s - 1)\cosh(\gamma d)}{\sinh(\gamma d) + (z_s + 1)\cosh(\gamma d)}$$

Output-residual gain G = 5.043

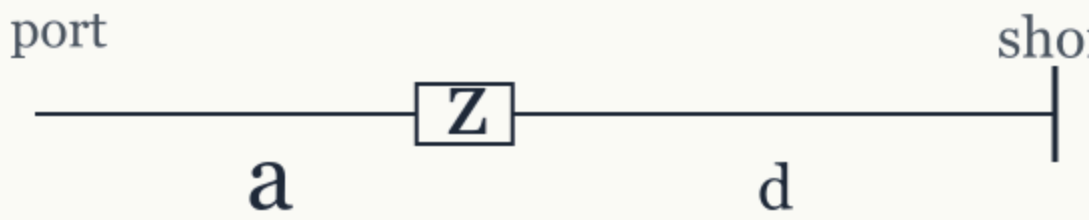
$\Delta\varphi = 179.686°$

The common factor rotates and attenuates.

The response and measured gain are non-monotonic in geometry.

Does the conventional solution converge?

Same finite switch and boundary conditions. Independent wave and ABCD checks agree.



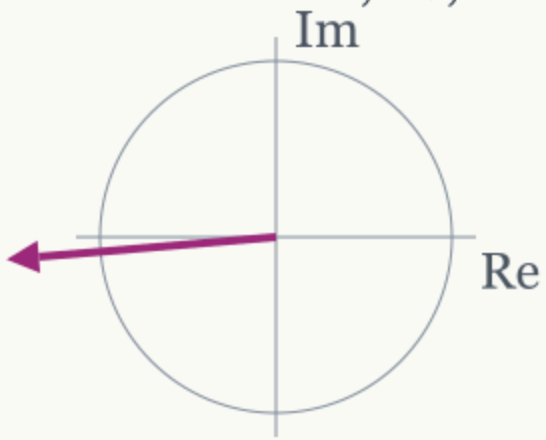
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



Complex reflection error, ×8,000



Refine the spatial solve

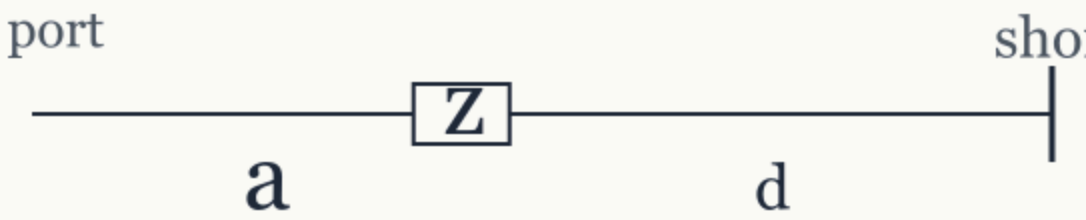
16 intervals / segment

$|\Delta R| = 0.0169043$

Doubling resolution shrinks the error.

Does the conventional solution converge?

Same finite switch and boundary conditions. Independent wave and ABCD checks agree.



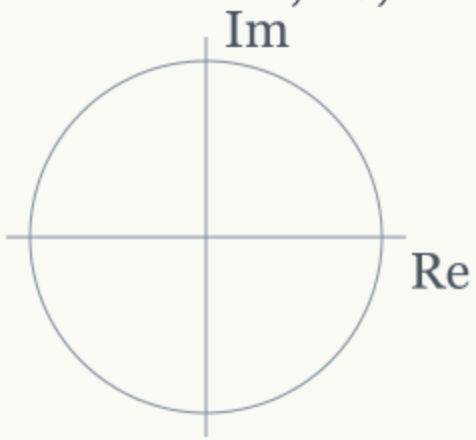
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



Complex reflection error, ×8,000



Refine the spatial solve

512 intervals / segment

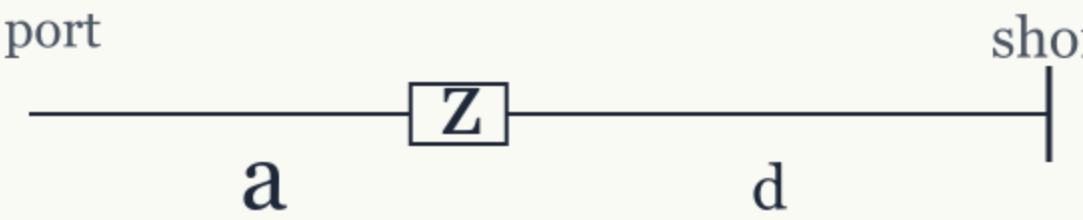
$|\Delta R| = 0.0000166$

Doubling resolution shrinks the error.

The spatial baseline converges at second order; power checks add independent evidence.

How does a local residual enter the loss?

Saved primary PINN seed 0, OFF, d = 0.208395. Physics-only training contains no field labels.



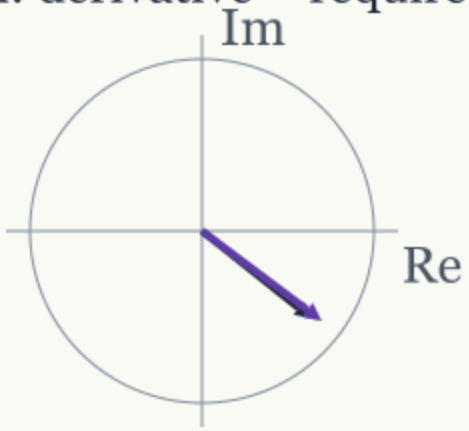
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



V equation: derivative – required



$\xi = 0.15$

$$L = \langle |f_V|^2 \rangle + \langle |f_I|^2 \rangle + \frac{1}{4}\langle |b_0|^2 \rangle + \langle |b_1|^2 \rangle$$

fV = -0.035 +0.007i

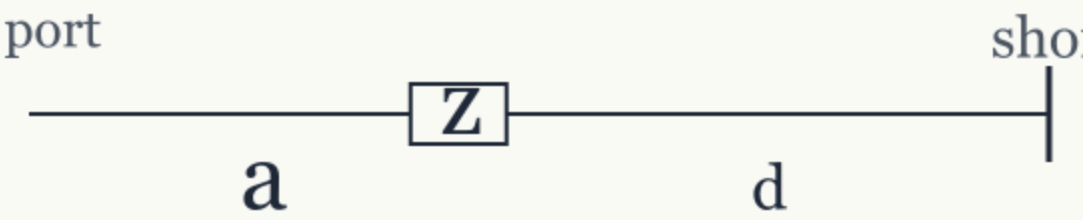
fI = -0.013 -0.159i

Integrated interior = 0.00000

Boundary contribution = 0.06148

How does a local residual enter the loss?

Saved primary PINN seed 0, OFF, d = 0.208395. Physics-only training contains no field labels.



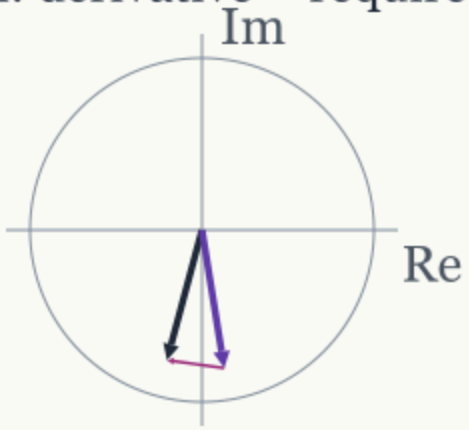
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



V equation: derivative – required



$\xi = 0.85$

$$L = \langle |f_V|^2 \rangle + \langle |f_I|^2 \rangle + \frac{1}{4}\langle |b_0|^2 \rangle + \langle |b_1|^2 \rangle$$

fV = -0.205 +0.028i

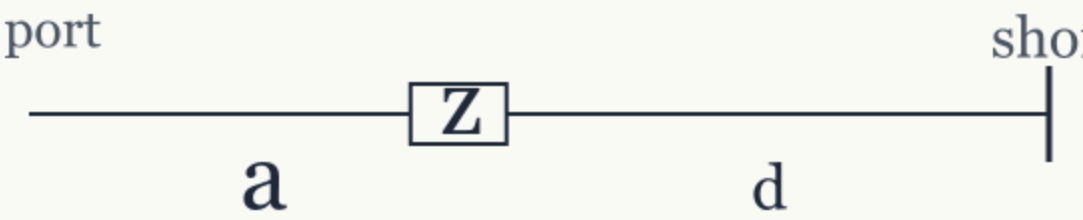
fI = -0.032 +0.036i

Integrated interior = 0.03291

Boundary contribution = 0.06148

How does a local residual enter the loss?

Saved primary PINN seed 0, OFF, d = 0.208395. Physics-only training contains no field labels.



$L = a + d$

L=0.360 a=0.152 d=0.208

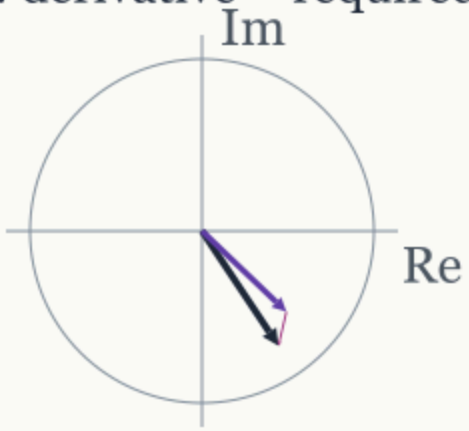
Port reflection

ON

OFF



I equation: derivative – required



$\xi = 0.35$

$$L = \langle |f_V|^2 \rangle + \langle |f_I|^2 \rangle + \frac{1}{4}\langle |b_0|^2 \rangle + \langle |b_1|^2 \rangle$$

fV = -0.061 +0.038i

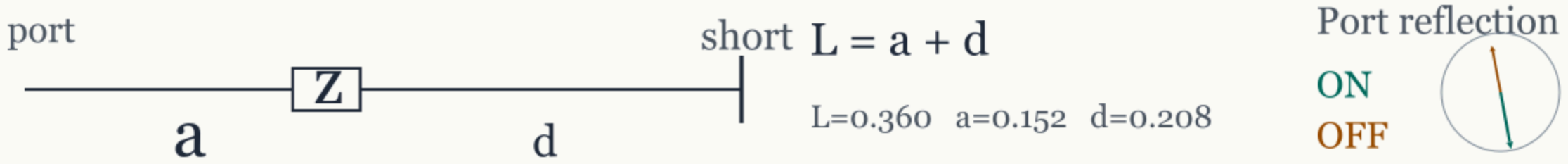
fI = -0.028 -0.121i

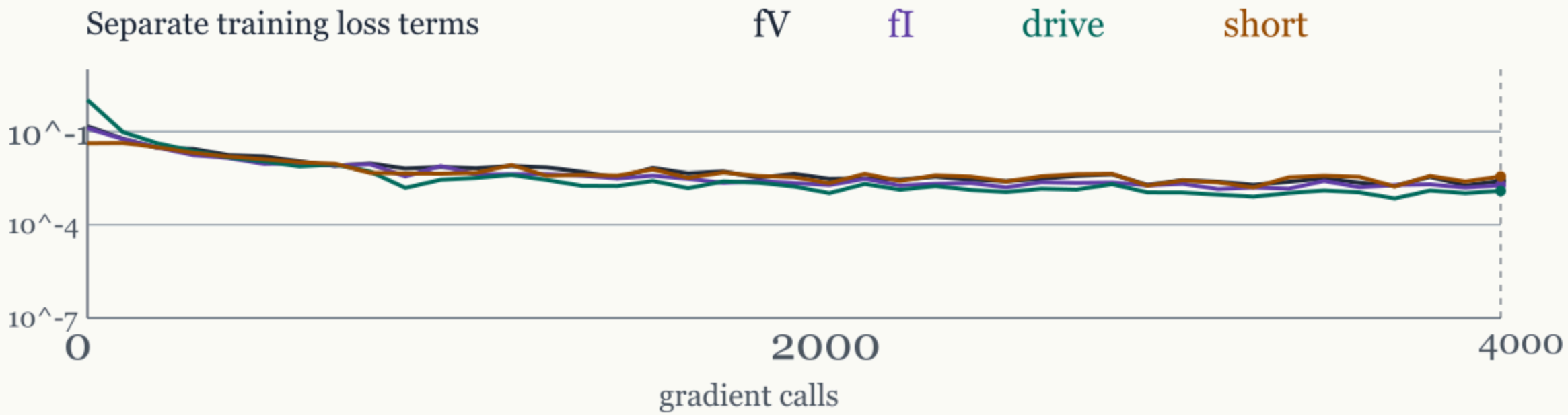
Integrated interior = 0.03291

Boundary contribution = 0.06148

How does a local residual enter the loss?

Saved primary PINN seed 0, OFF, d = 0.208395. Physics-only training contains no field labels.

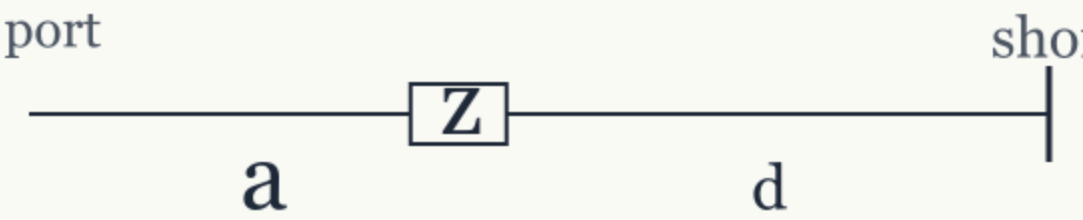




Small sampled loss and exact endpoints still need field and output checks.

What can the real projection hide?

Example: seed 0, OFF, d = 0.208395. Axes are ξ, Re V, Im V, not device geometry.

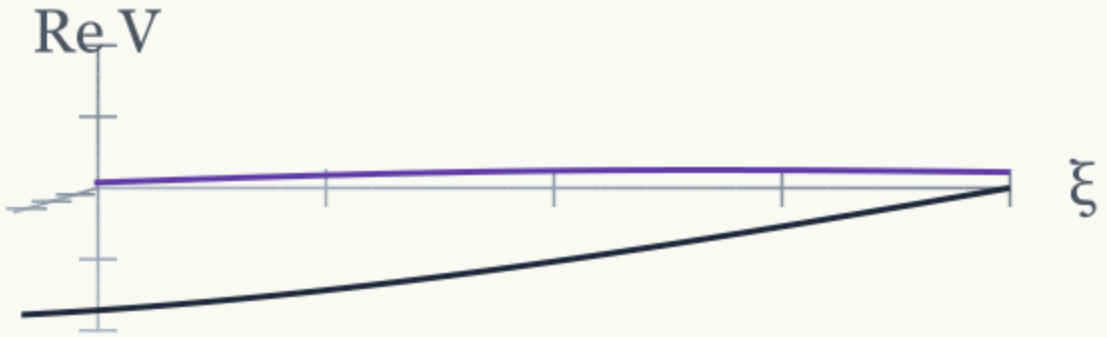


$L = a + d$

L=0.360 a=0.152 d=0.208

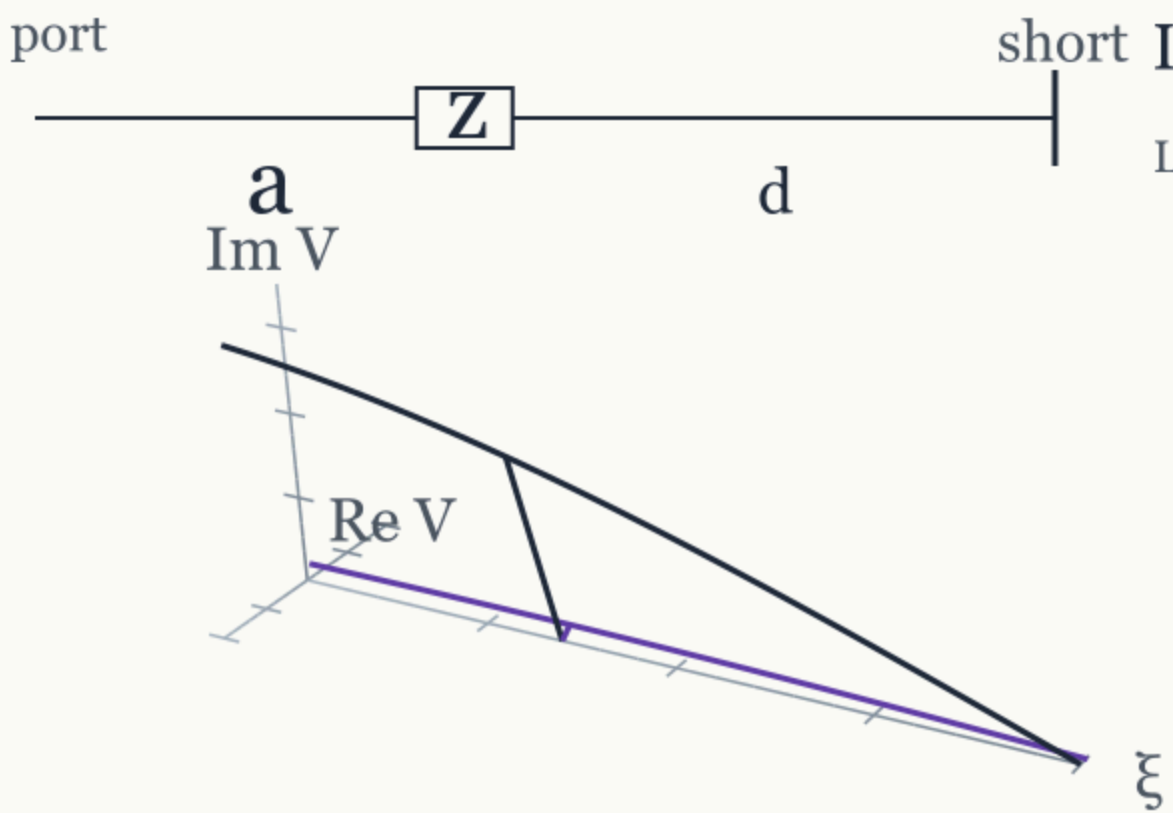


learned reference



What can the real projection hide?

Example: seed 0, OFF, d = 0.208395. Axes are ξ, Re V, Im V, not device geometry.



$L = a + d$

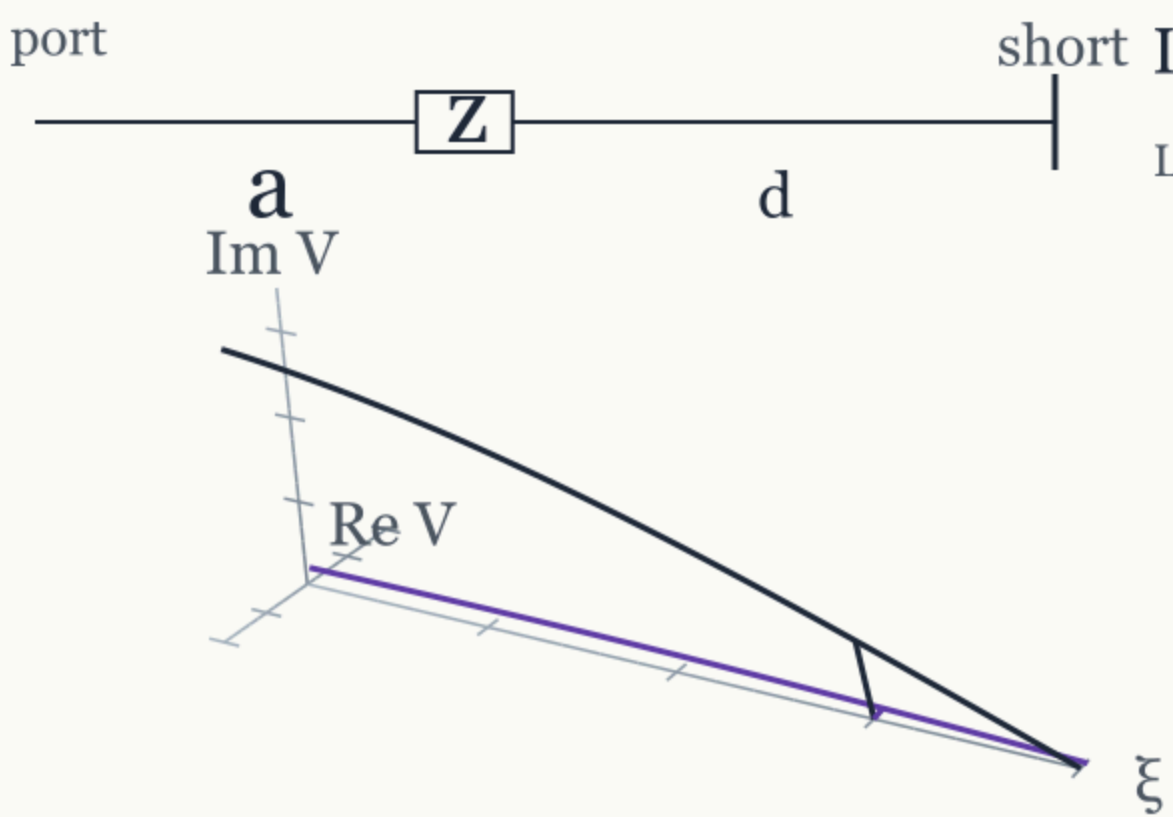
L=0.360 a=0.152 d=0.208



learned reference

What can the real projection hide?

Example: seed 0, OFF, d = 0.208395. Axes are ξ, Re V, Im V, not device geometry.



L = a + d

L=0.360 a=0.152 d=0.208



learned reference

A smooth projection can hide a large complex-field disagreement.

Which spatial components remain in the error?

Complex voltage error at one fixed geometry. DFT uses 128 samples, excluding the duplicate endpoint.



$L = a + d$

L=0.360 a=0.152 d=0.208

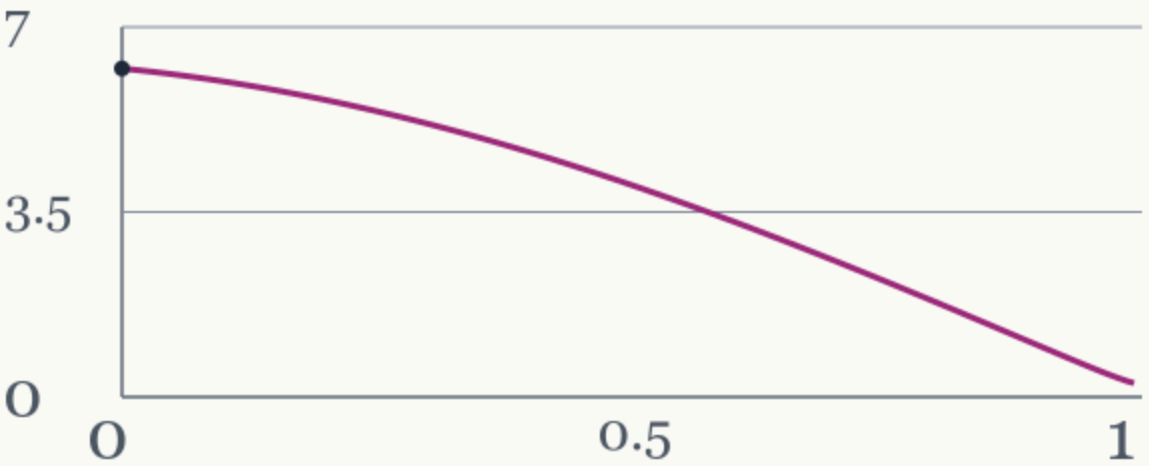
Port reflection

ON

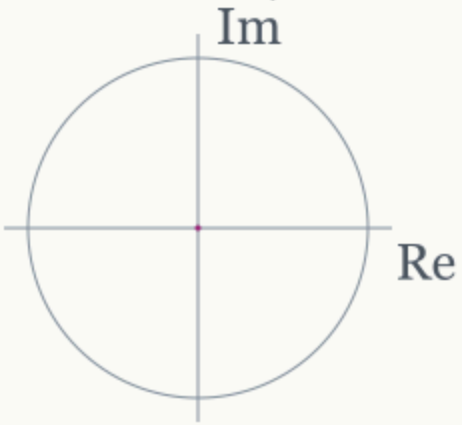
OFF



Absolute complex V error



Weighted Fourier sum / 128



k = 0 samples = 0/128

Which spatial components remain in the error?

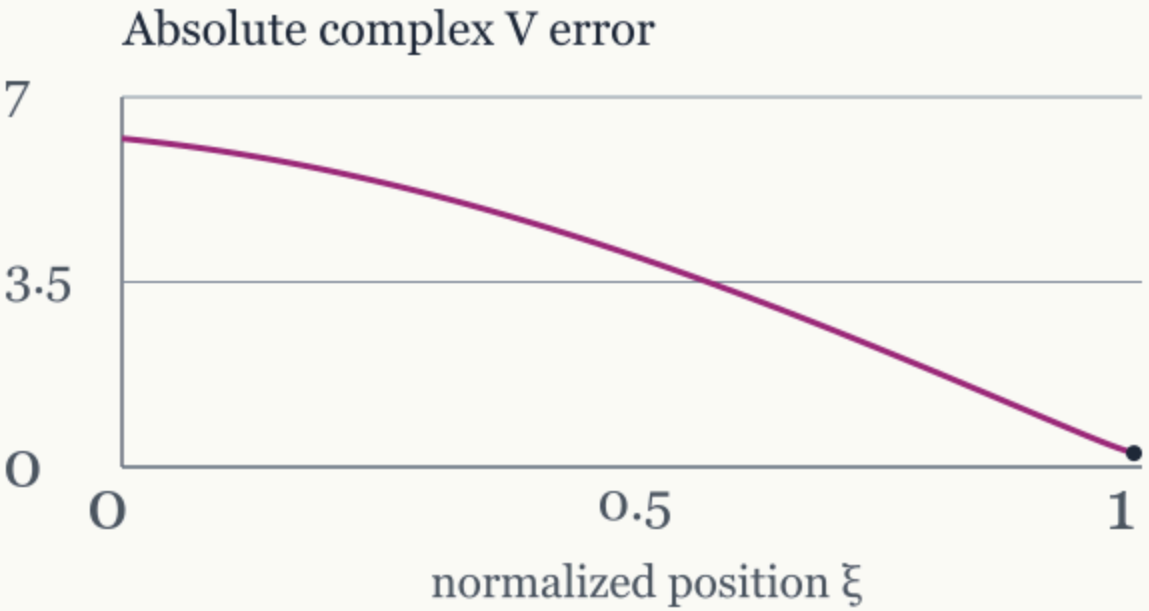
Complex voltage error at one fixed geometry. DFT uses 128 samples, excluding the duplicate endpoint.

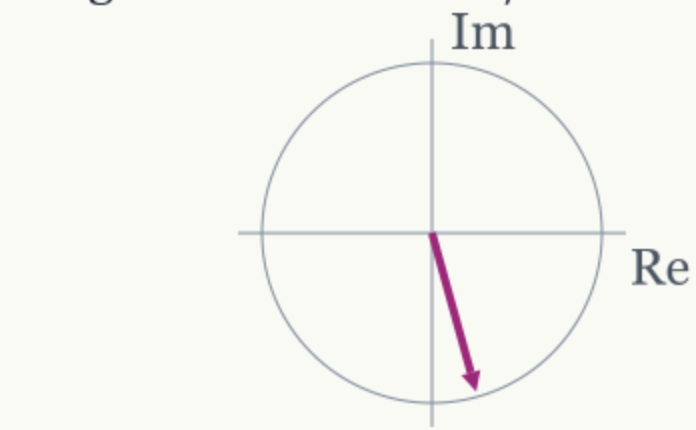


$L = a + d$

L=0.360 a=0.152 d=0.208







k = 0 samples = 128/128

Which spatial components remain in the error?

Complex voltage error at one fixed geometry. DFT uses 128 samples, excluding the duplicate endpoint.



$L = a + d$

L=0.360 a=0.152 d=0.208

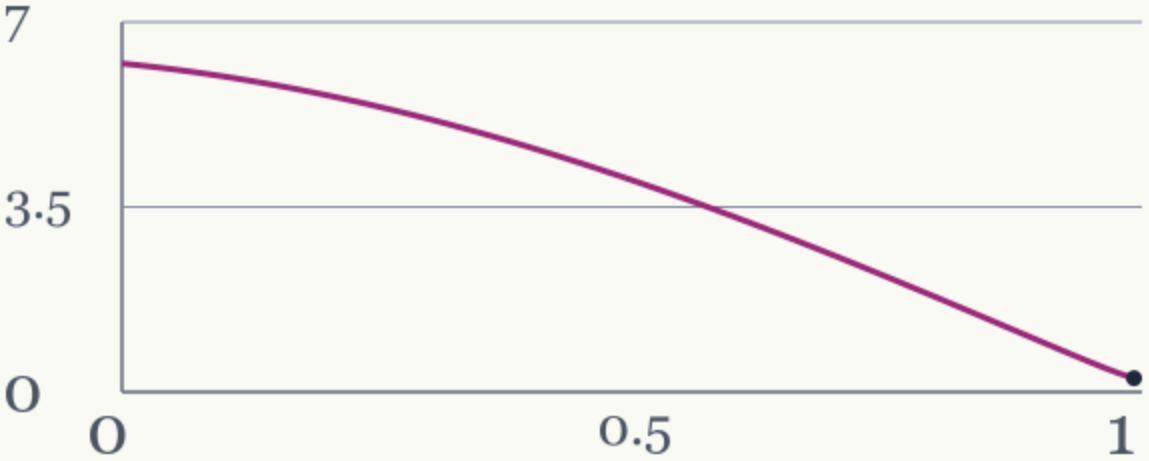
Port reflection

ON

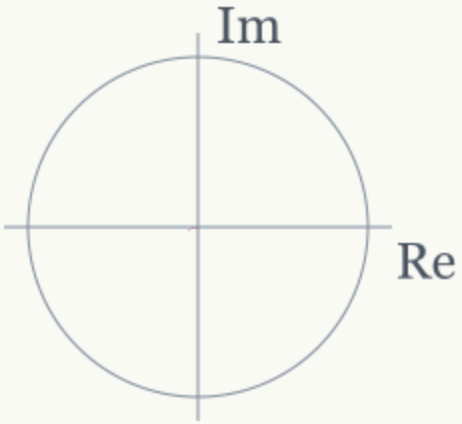
OFF



Absolute complex V error



Weighted Fourier sum / 128



k = 4 samples = 128/128

Which spatial components remain in the error?

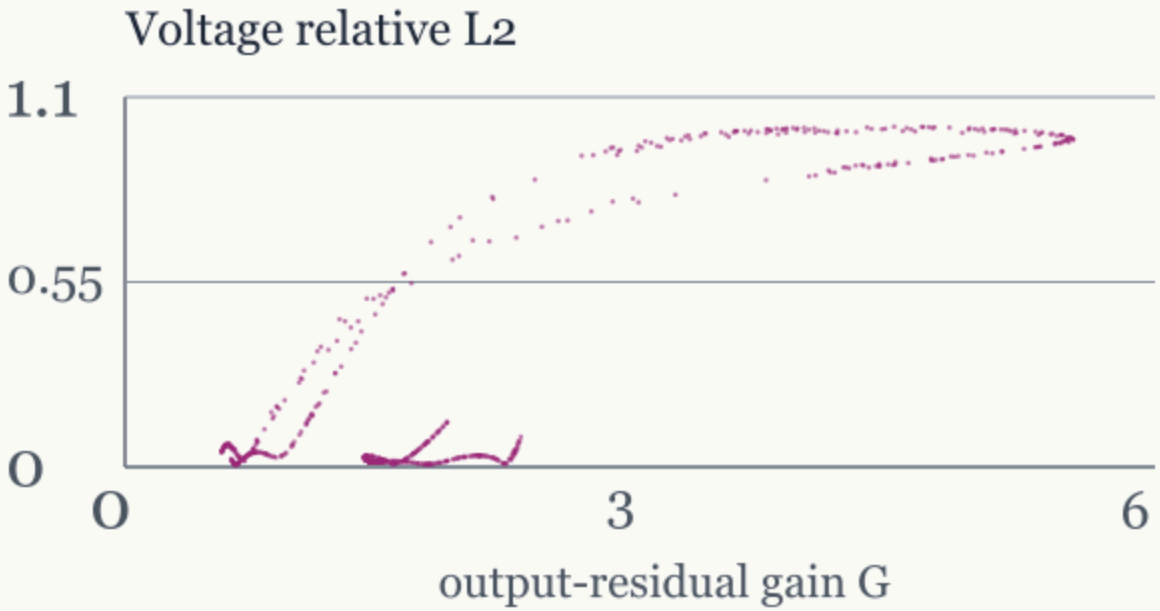
Left: held-out state errors versus measured gain. Right: the fixed-geometry spatial spectrum.

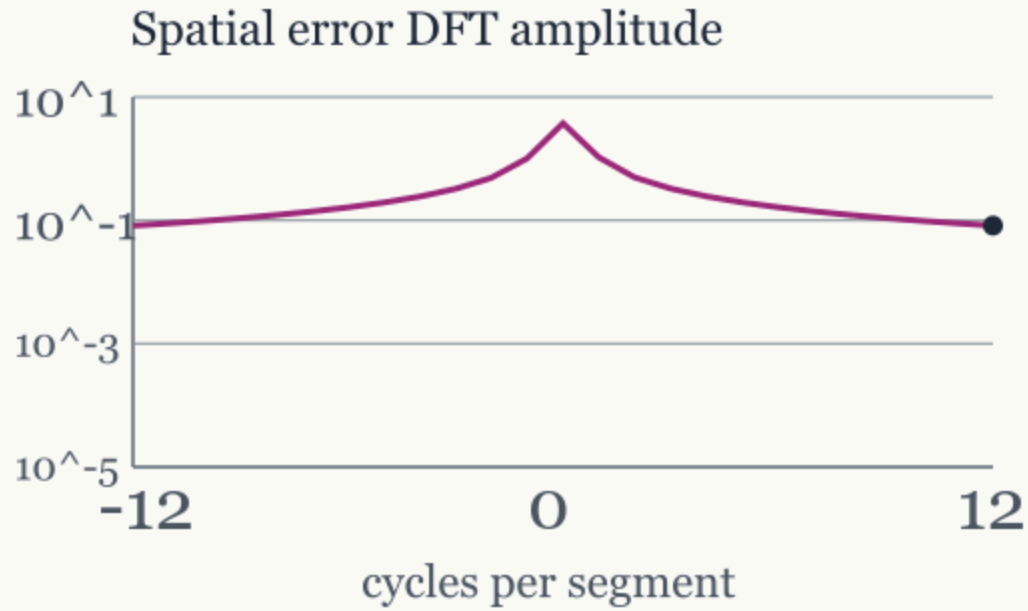


$L = a + d$

L=0.360 a=0.152 d=0.208



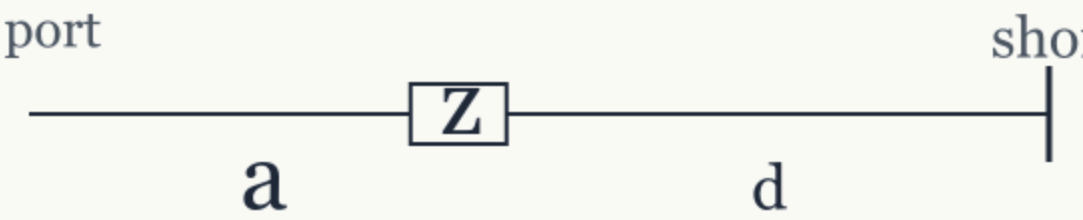




Gain and spectra locate errors; they do not identify the training-failure cause.

How does residual error reach the reflected wave?

Actual saved checkpoint and geometry. The verified identity transports output error, not training causes.



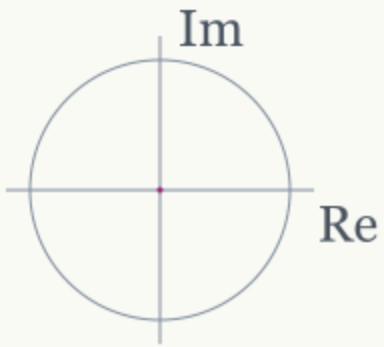
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



$$\Delta R = \frac{-e^{2\gamma a} c\, b_0}{\underset{\sim}{D}}_{\text{drive}} + \frac{-e^{-2\gamma a} q}{\underset{\sim}{D}}_{\text{interior}} + \frac{e^{2\gamma a} b_1}{\underset{\sim}{D}}_{\text{short}}$$



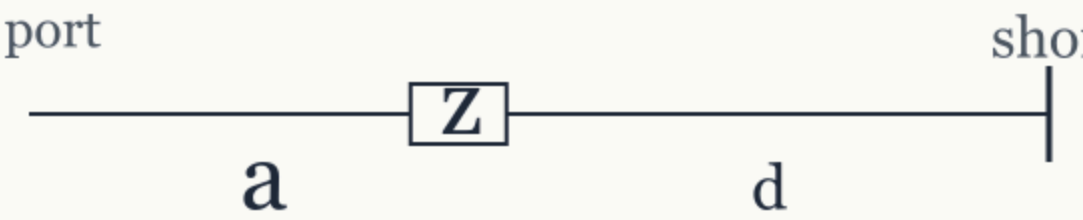
$\xi = 0.00$

$q = +0.000 + 0.000i$

Divide by complex D: rotate and scale

How does residual error reach the reflected wave?

Actual saved checkpoint and geometry. The verified identity transports output error, not training causes.



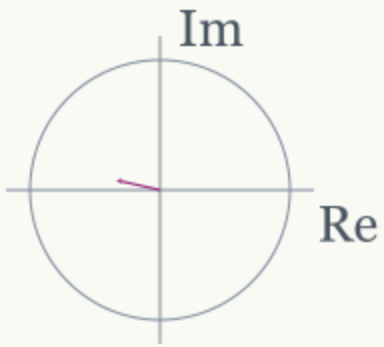
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



$$\Delta R = \frac{-e^{2\gamma a} c\, b_0}{\underset{\sim}{D}_{\text{drive}}} + \frac{-e^{-2\gamma a} q}{\underset{\sim}{D}_{\text{interior}}} + \frac{e^{2\gamma a} b_1}{\underset{\sim}{D}_{\text{short}}}$$



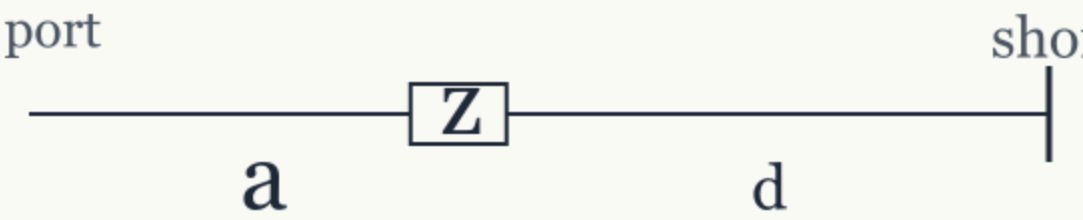
$\xi = 1.00$

$q = -0.168 + 0.038i$

Divide by complex D: rotate and scale

How does residual error reach the reflected wave?

Actual saved checkpoint and geometry. The verified identity transports output error, not training causes.



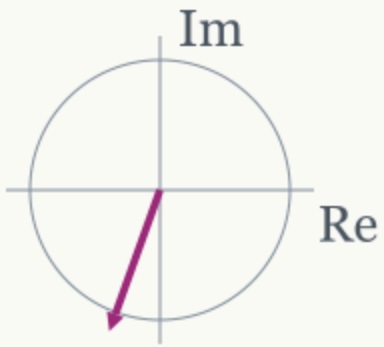
$L = a + d$

L=0.360 a=0.152 d=0.208

Port reflection



$$\Delta R = \underbrace{\frac{-e^{2\gamma a} c\, b_0}{D}}_{\text{drive}} + \underbrace{\frac{-e^{-2\gamma a} q}{D}}_{\text{interior}} + \underbrace{\frac{e^{2\gamma a} b_1}{D}}_{\text{short}}$$



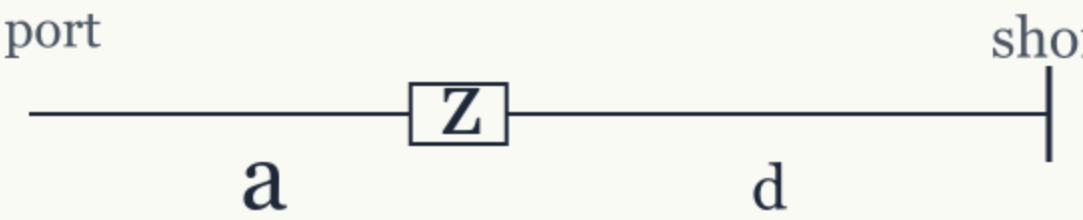
$\xi = 1.00$

$q = -0.168 + 0.038i$

Divide by complex D: rotate and scale

How does residual error reach the reflected wave?

Actual saved checkpoint and geometry. The verified identity transports output error, not training causes.



$L = a + d$

L=0.360 a=0.152 d=0.208

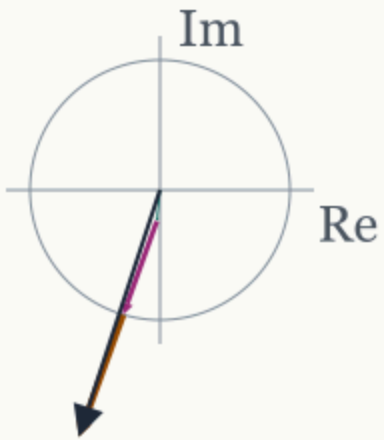
Port reflection

ON

OFF



$$\Delta R = \frac{-e^{2\gamma a} c\, b_0}{\underset{\text{drive}}{D}} + \frac{-e^{-2\gamma a} q}{\underset{\text{interior}}{D}} + \frac{e^{2\gamma a} b_1}{\underset{\text{short}}{D}}$$



thin dark = measured ΔR

The signed contributions can cancel.

Their actual complex sum matches predicted minus reference reflection.

Does enforcing the endpoints repair the model?

One trained hard-model raw field is transformed in stages. The separate paired experiment used six fresh fits.



$L = a + d$

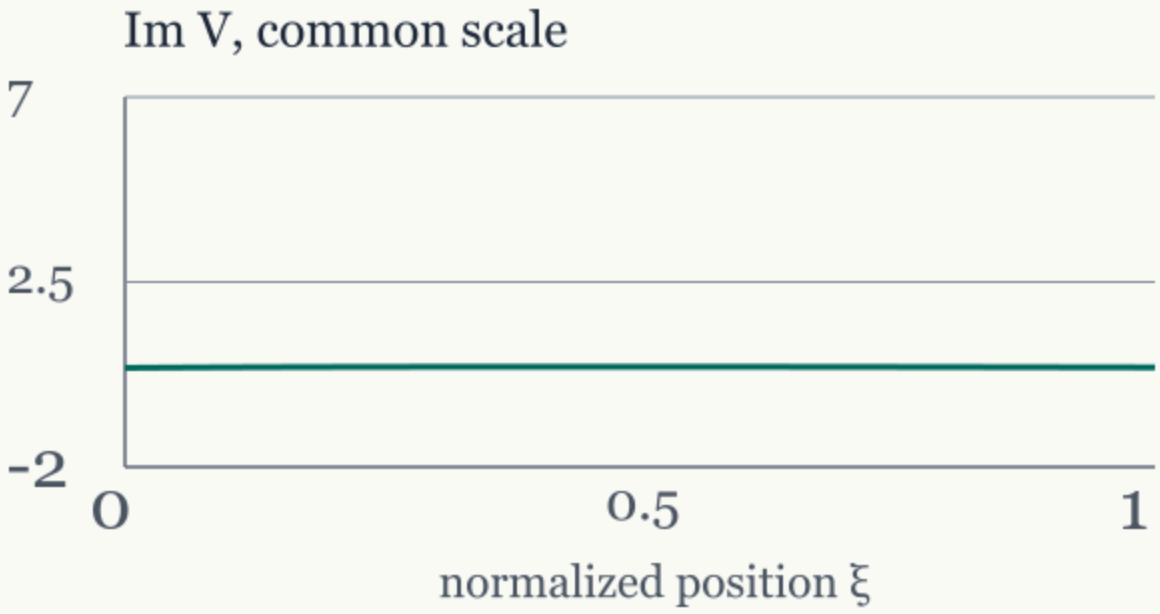
L=0.360 a=0.152 d=0.208

Port reflection

ON

OFF





$$I = i, \qquad V = v - \xi v(1) - (1 - \xi)[v(0) + (1 + z_s)i(0) - 2]$$

|drive residual| = 1.31221

|V(1)| = 0.95349

Does enforcing the endpoints repair the model?

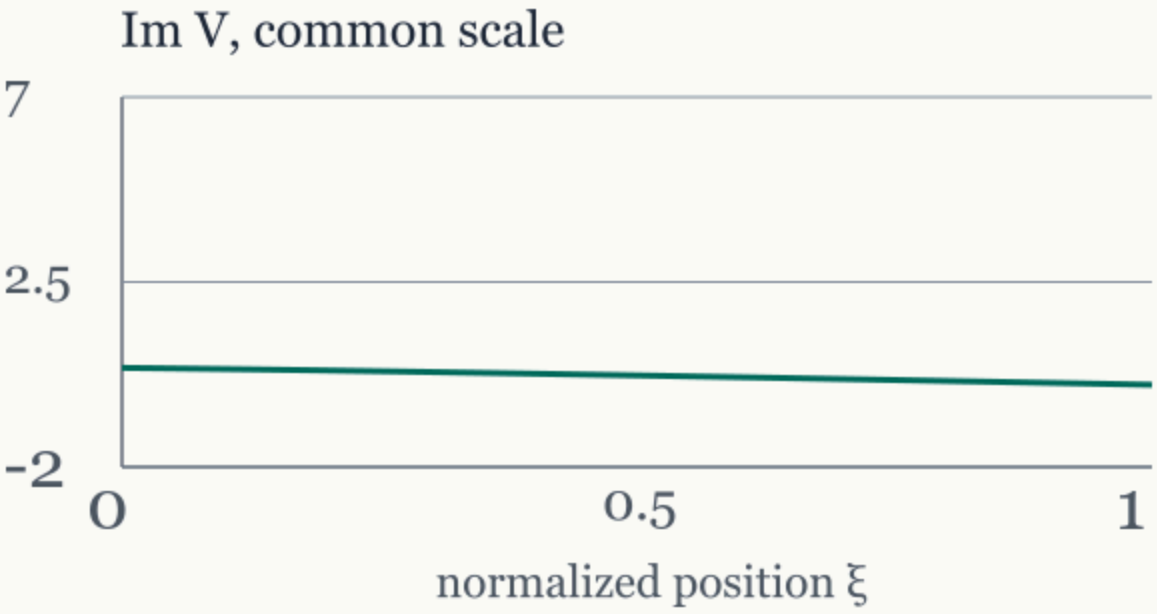
One trained hard-model raw field is transformed in stages. The separate paired experiment used six fresh fits.



$L = a + d$

L=0.360 a=0.152 d=0.208





$$I = i, \qquad V = v - \xi v(1) - (1-\xi)[v(0) + (1+z_s)i(0) - 2]$$

|drive residual| = 1.31221

|V(1)| = 0.00000

Does enforcing the endpoints repair the model?

One trained hard-model raw field is transformed in stages. The separate paired experiment used six fresh fits.



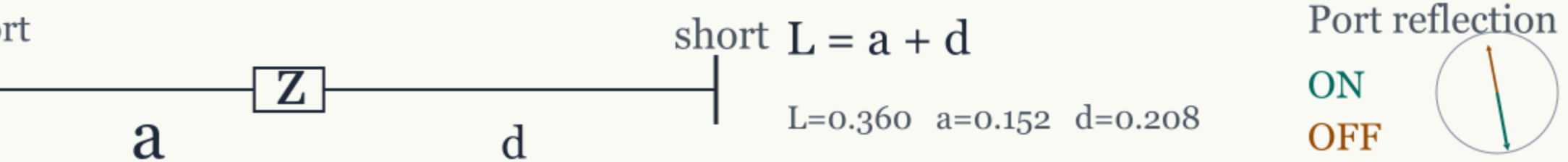
$L = a + d$

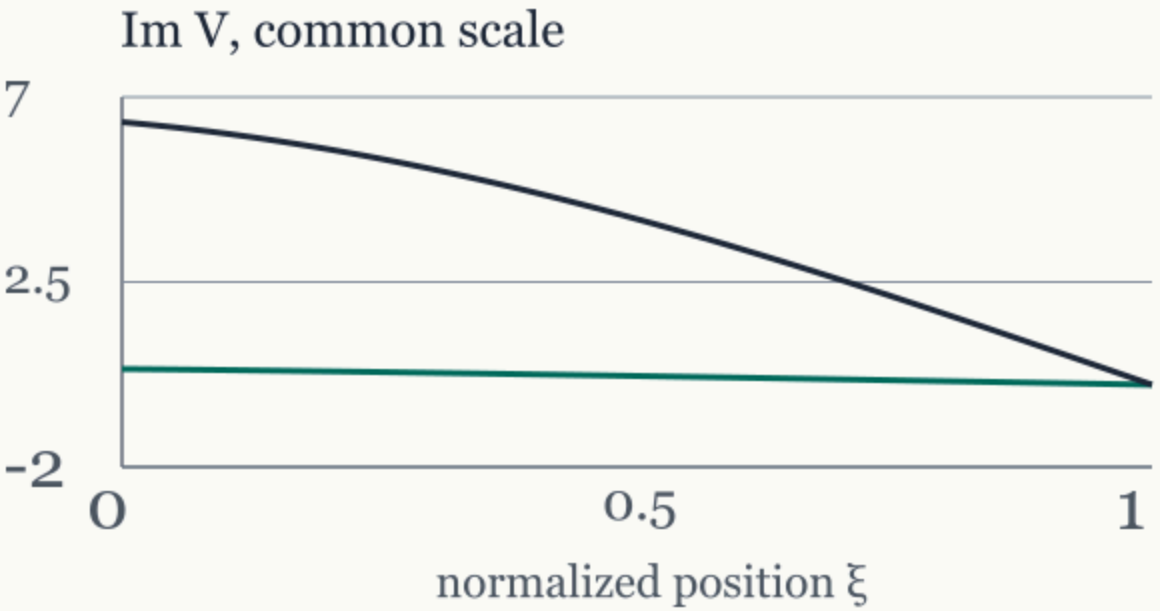
L=0.360 a=0.152 d=0.208

Port reflection

ON

OFF





$$I = i, \qquad V = v - \xi v(1) - (1 - \xi)[v(0) + (1 + z_s)i(0) - 2]$$

|drive residual| = 0.00000

|V(1)| = 0.00000

Does enforcing the endpoints repair the model?

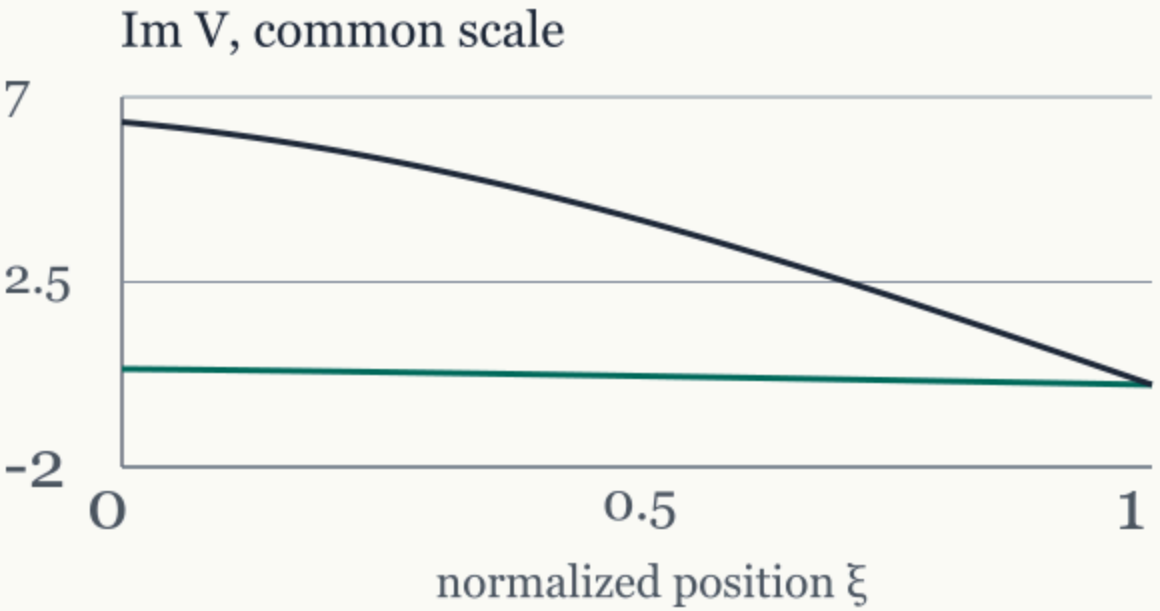
One trained hard-model raw field is transformed in stages. The separate paired experiment used six fresh fits.



$L = a + d$

L=0.360 a=0.152 d=0.208





$$I = i, \qquad V = v - \xi v(1) - (1-\xi)[v(0) + (1+z_s)i(0) - 2]$$

|drive residual| = 0.00000

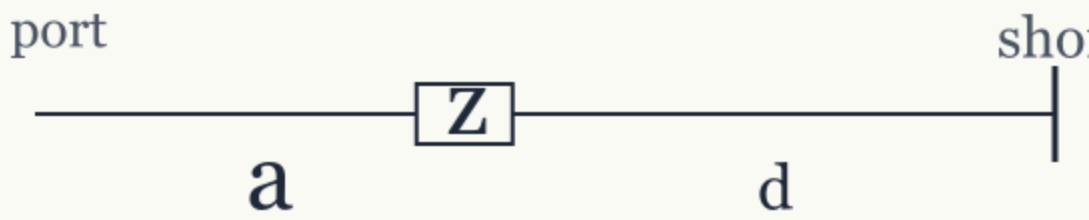
|V(1)| = 0.00000

All six validation gates failed.

Exact endpoint enforcement was insufficient to repair this model at the fixed budget.

What do controlled comparisons establish?

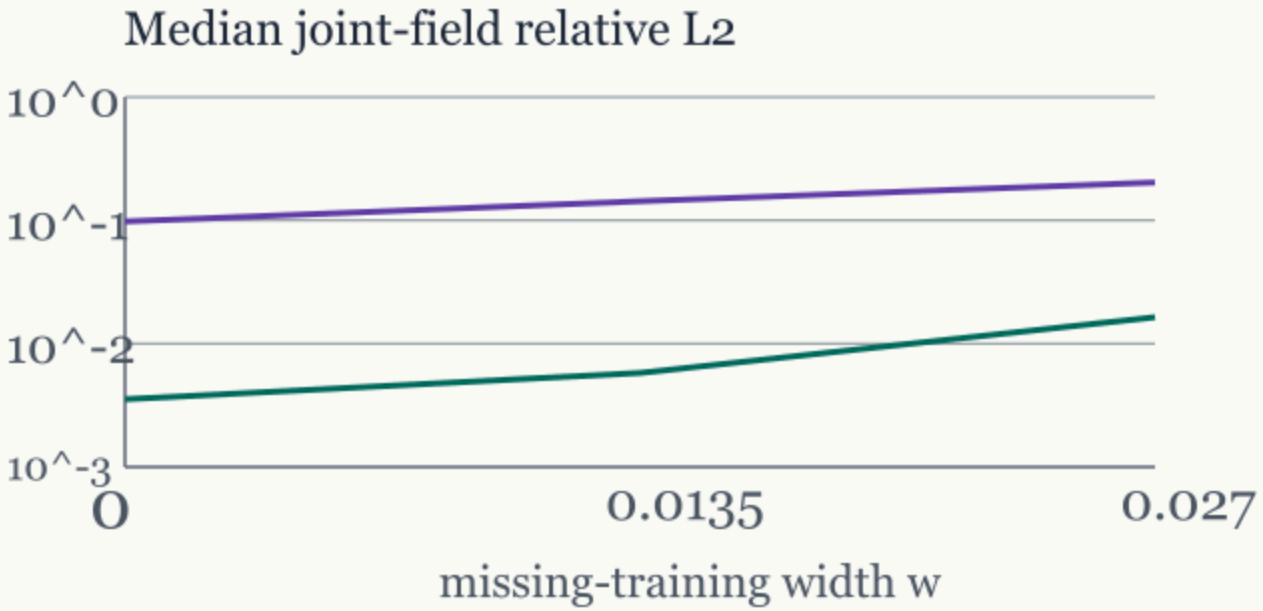
P3: increase the missing-training width w around d = 0.2085. Equations stay fixed.



$L = a + d$

L=0.360 a=0.152 d=0.208





physics objective

direct-reference objective

One shared primary warm start per pair

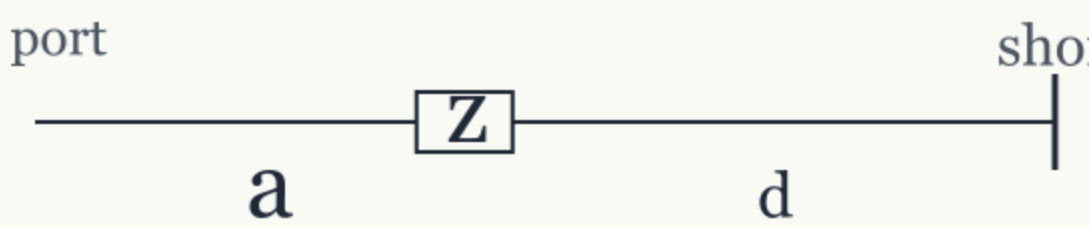
Same network, inputs, L-BFGS, 1,200 calls

Training counts: 256 / 248 / 240

One seed; historical evaluation cohorts

What do controlled comparisons establish?

Same six fits: diagnostic improvement is separate from the unchanged deployment gate.



$L = a + d$

L=0.360 a=0.152 d=0.208



27.63x

lower median field error at w = 0

Change physics to reference objective

4.63x

higher reference-fit error at largest gap

All six BLOCKED gates failed

Three supervised OFF-GRID gates passed

Full coverage: retains 10/10 feasible

but promotes 10/150 infeasible designs

R95 = 0.04135 exceeds 0.03

Nested removal changes N, not fixed-N location.

What do controlled comparisons establish?

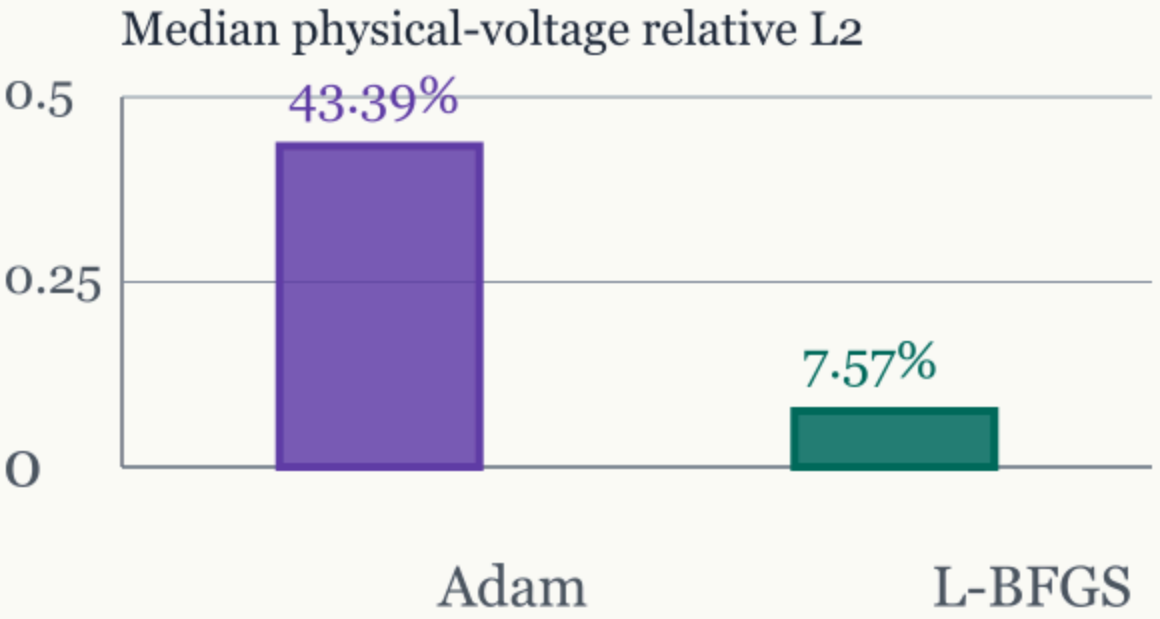
Separate optimizer control: covered checkpoint, physical two-segment voltage metric; do not mix metrics.



$L = a + d$

L=0.360 a=0.152 d=0.208





Same start, physics objective and inputs

Same architecture and 1,200 gradient calls

204/256 blocked state errors improve

Both arms retain 0/8 feasible designs

Both validation gates fail

Conditional effect; one seed, not a unique cause.

Optimizer, objective and support affect attained error. No Path C success is claimed.

Would this support useful, cheaper design screening?

Illustrative requirements: phase ≥165°, minimum magnitude ≥0.8. They are not the researcher's stated tolerances.



$L = a + d$

L=0.360 a=0.110 d=0.250



Reference decision

$\Delta\varphi = 0.026°$

min |R| = 0.9786

INFEASIBLE

Saved neural decision

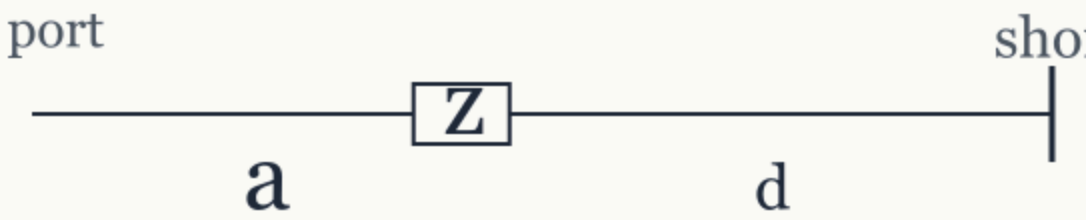
$\Delta\varphi = 15.215°$

min |R| = 0.9311

INFEASIBLE

Would this support useful, cheaper design screening?

Illustrative requirements: phase ≥165°, minimum magnitude ≥0.8. They are not the researcher's stated tolerances.



$L = a + d$

L=0.360 a=0.152 d=0.208



Reference decision

$\Delta\varphi = 179.686°$

min |R| = 0.8009

FEASIBLE

Saved neural decision

$\Delta\varphi = 54.874°$

min |R| = 0.8928

INFEASIBLE

Would this support useful, cheaper design screening?

Illustrative requirements: phase ≥165°, minimum magnitude ≥0.8. They are not the researcher's stated tolerances.



$L = a + d$

L=0.360 a=0.152 d=0.208



Primary held-out design cohorts

Off-grid: 21/21 feasible retained

Blocked: 0/10 feasible retained

Same blocked failure in all three PINN seeds.

Original thresholds and saved cohorts.

Fresh paired boundary control

All six validation gates failed.

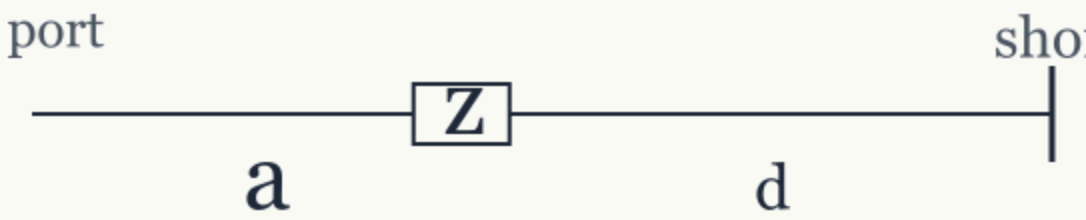
Blocked final: 30 infeasible cases

No feasible cases; sensitivity undefined.

Historical cohorts remain diagnostic.

Would this support useful, cheaper design screening?

Illustrative requirements: phase ≥165°, minimum magnitude ≥0.8. They are not the researcher's stated tolerances.



$$L = a + d$$

L=0.360 a=0.152 d=0.208



$$T_{\text{total}}(N) = T_{\text{setup}} + T_{\text{train}} + NT_{\text{inf}}$$

10,000,000 two-state queries

Conventional: 117.2500 s

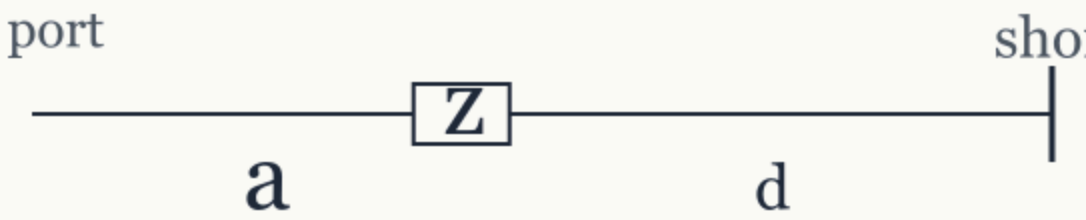
Neural + fit: 1914.5875 s

CPU float64, one thread. One-fit lower bound; full setup/history is unmeasured.

There is no positive break-even here, and the neural accuracy gate fails.

What should the researcher do next?

A bounded synthetic result with a concrete next researcher decision.



$L = a + d$

L=0.360 a=0.152 d=0.208



Established in this analogue

Finite-switch influence depends on participation.

Exact endpoints alone did not repair the model.

Needed for hardware transfer

One permitted unit-cell family and switch model

paired responses and useful tolerances

solver, storage and orchestration cost separately

Keep the conventional reference. Validate this mechanism before claiming device benefit.

Sources for the physical and diagnostic argument

Full derivations, source hashes, frozen protocols and uncertainty markers are retained in the report.



$L = a + d$

L=0.360 a=0.152 d=0.208



Team interview brief and recording

Approved source copies linked from the report

Chew: lossy transmission lines

wcchew.ece.illinois.edu/chew/ece350/ee350-11.pdf

Costa & Borgese, 2021

doi.org/10.1109/OJCOMS.2021.3092217

Krishnapriyan et al., 2021

arxiv.org/abs/2109.01050

Sukumar & Srivastava, 2021

arxiv.org/abs/2104.08426

Grossmann et al., 2024

doi.org/10.1093/imamat/hxae011